
Decision-making Model for Combined Transportation System of Lunar Colonies Based on Pareto Optimization

Summary

As lunar colonization programs transition into the phase of substantive construction, the development of an efficient, economical, and environmentally sustainable Earth-Moon logistics framework has emerged as a critical challenge. Aiming at the objective of establishing a lunar colony for 100,000 residents by 2050, this paper constructs a series of multi-dimensional coupled mathematical models to address the transport of 100 million tons of infrastructure materials, adaptation to non-ideal operating conditions, annual water resource replenishment, and Earth's atmospheric environmental constraints.

Regarding **Problem I**, a Hybrid Capacity Trade-off Optimization model based on Pareto optimality (HCTO-PM) was developed. By incorporating Wright's Law, the model establishes a dynamic cost iteration mechanism to quantitatively compare the space elevator (Option A), traditional rockets (Option B), and hybrid transportation (Option C). The results indicate that the hybrid approach performs optimally on the "time-cost" Pareto frontier: under a balanced decision-making strategy, completing the infrastructure mission requires 77 years at a total cost of approximately \$11.5 trillion.

Regarding **Problem II**, a fault quantification and correction model was established. Considering stochastic factors such as cable oscillations, equipment failure rates, and rocket launch failure rates (5%), the Pareto frontier was recalculated. Sensitivity analysis based on the single-parameter local perturbation method demonstrates that the pure elevator scheme exhibits the highest sensitivity to schedule fluctuations (with a 30.7% duration extension), whereas the hybrid scheme demonstrates superior engineering resilience and risk mitigation through the synergistic effect of "fixed orbits plus discrete backups."

Regarding **Problem III**, a Dynamic Mass Balance (DMS) model was constructed. By integrating data from human metabolism, Controlled Ecological Life Support Systems (BLSS), industrial oxygen production, and technical leakage, the net annual water resource deficit for a 100,000-person colony was calculated to be 35.4 kilotonnes. A comparison of replenishment pathways identifies the "elevator-primary with rocket-elastic-backup" mode as the optimal strategy, which reduces annual operating expenditures by approximately 14.2% compared to pure rocket replenishment.

Regarding **Problem IV**, this paper defines an Environmental Impact Unit (EIU) based on the Life Cycle Assessment (LCA) methodology. Utilizing the theory of stratospheric aerosol steady-state loading, it physically derives a threshold of 4,000 annual rocket launches as the maximum safety limit for Earth's atmosphere. By embedding this hard constraint into the optimization model, a three-dimensional Pareto mapping of "Time-Cost-Environment" was generated. The research proves that maximizing the utilization of space elevators is the only viable path for Earth's ecosystem to sustain the transport of 100 million tons of materials between the Earth and the Moon.

Finally, a comprehensive policy recommendation letter was submitted to the MCM organization, covering site selection strategies, In-Situ Resource Utilization (ISRU), and dynamic scheduling mechanisms.

Keywords: Pareto multi-objective optimization; Wright's Law; Life Cycle Assessment; Space Elevator

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1 Introduction

1.1 Background

As the earth's resources become increasingly scarce and the population continues to grow, human civilization is facing unprecedented sustainable development challenges. As the closest celestial body to the earth, the moon is rich in helium-3, titanium, iron and other mineral resources, as well as its strategic value as a springboard for deep space exploration. It is a key target for international space agencies to compete for layout.

However, traditional space transportation systems using rocket transportation have significant bottlenecks. Currently, according to SpaceX data, the cost of sending a 1 kilogram payload into low-Earth orbit is about US\$2,720, and the cost of reaching the lunar surface is even higher. This high-cost, low-frequency transportation mode severely restricts the construction of large-scale space infrastructure. At the same time, this solution also seriously affects the earth's ecological environment due to the harmful gas emissions produced by the fuel.

In contrast, the space elevator system has shown breakthrough potential as a revolutionary space transportation infrastructure. The system uses the balance between the centrifugal force of the Earth's rotation and gravity to set a vertex anchor point in the geosynchronous orbit 35,786 kilometers above the equator, and connects to ground ports through a 100,000-kilometer-long super-strong material tether to form a "space highway." Compared with traditional rockets, space elevators use electricity as a power source and can theoretically achieve large-scale material transportation with nearly zero carbon emissions, reducing unit transportation costs to less than one thousandth of traditional methods.

This topic focuses on the envisaged plan to colonize the moon with 100,000 people in 2050. It analyzes the time cycle and economic cost of transportation of engineering materials in the early stages of lunar base construction, and the different transportation options that can be adopted for the water demand of the base after residents move in. Then, it comprehensively considers the environmental impact of different transportation options and provides different transportation options.

1.2 Restatement of the Problem

Problem1: It is estimated that the construction of a lunar colony of 100,000 people will require 100 million tons of materials. Under ideal conditions (no rocket launch failure, no tether shaking), three different options are adopted:

- a: Three galactic ports using only space elevators
- b: Traditional solution using only ten Earth rocket launch systems
- c: A combination of two transport systems

This article will establish a mathematical model to calculate the transportation costs and time required for these three different options.

Problem2: Expand the model of Question 1 to consider the impact of non-ideal factors on the transportation plan, that is, consider factors such as cable shaking of the space elevator system, elevator damage, and launch failure factors of the rocket transportation system. Due to these non-ideal factors, the annual carrying capacity of both transportation systems, as well as the unit transportation costs, will be affected. These effects are combined in the model of Question 1 to obtain the adjusted transportation plan.

Problem3: After the lunar colony is completed and operated, what is the total water demand for a population of 100,000 and its distribution? A mathematical model is established to analyze this problem. Combined with the transportation system model of questions 1 and 2, a water delivery plan with optimal additional costs and timing is obtained on the premise of ensuring water supply for the colony.

Problem 4: During the process of transporting materials, launching rockets will inevitably have an impact on the earth's environment. However, the earth's environment can only bear limited pollution. This impact will be taken into consideration in the model to re-formulate the material and water resources transportation plan for the realization of a 100,000-person lunar colony base.

1.3 The Description of the Problem

Problem1: The scene in question 1 is set against the background of the initial construction of the lunar colony. Humanity has built a complete space elevator system, which can transport supplies to the moon through the Galaxy Port, and can also use ground rocket launch systems to transport supplies to the lunar base. Under the assumption that both the rocket launch and the space elevator system work under ideal conditions (no capacity loss, no additional cost), three transportation options are used: three galactic ports relying only on the space elevator system, traditional rocket launch using only existing bases, or a combination of the two methods. The time and economic costs required for different options are modeled and analyzed. Among them, in the combination scheme of the two methods, in order to solve the comprehensive optimization of time and economy and analyze the rocket launch arrangement, the multi-objective Pareto optimization model can be used for analysis. At the same time, considering the possible decline in rocket manufacturing costs, the annual decay coefficient of rocket manufacturing is introduced based on Wright's law, and a rocket cost iteration model is added to Plan C to optimize the plan.

Problem2: The ideal assumption of Question 1 ignores the inherent risks of the space transportation system: the ultra-long cable of the space elevator is affected by orbital disturbance and mechanical wear, and there are problems such as cable shaking and equipment failure; traditional rocket launches are affected by weather and technical complexity, and face risks such as launch failure and reduced launch site efficiency. Such non-ideal factors will directly lead to capacity attenuation, cost increase and construction period extension. It is necessary to quantify the impact of faults and correct the original model to ensure that the solution has reliability and anti-interference capabilities in actual engineering scenarios. Abstract faults can be converted into quantifiable coefficient indicators to build capacity attenuation and cost addition models. Embed non-ideal parameters into the multi-objective optimization model of Problem 1, recalculate the time and cost of the three types of options, generate the modified Pareto front, and obtain the adjustments of each option under non-ideal conditions.

Problem3: The total water demand after the 100,000-person lunar colony completes operations can be based on existing data from the International Space Station (ISS), Biosphere 2, etc., and a per capita daily water flux model can be constructed based on scenarios. At the same time, in order to maximize the utilization of water resources, a water recycling system must be considered. Then, the water demand, recycling efficiency and loss rate of each scenario are integrated to calculate the annual net water resource deficit (i.e. the annual water resource demand of the base). Embed the net supply demand into the early transportation model, compare the costs, capacity matching, and long-term operation and maintenance resilience of the three types of transportation routes, and introduce the redundant design idea of "main transportation + backup".

Problem 4: Large-scale Earth-to-Moon transportation will have a cumulative impact on the Earth's environment. Among them, the source of pollution can be considered to be only the ground rocket launch system, while the space elevator system has zero emissions and no pollution during transportation. First, the environmental cost of the transportation solution needs to be quantified. A composite environmental impact equivalent can be constructed based on the life cycle assessment (LCA) method and the environmental damage weight of multiple types of pollutants can be integrated. Secondly, the earth's environmental pollution carrying threshold can be derived. Based on the stratospheric aerosol steady-state load theory, the maximum annual number of rocket launches that the earth can withstand can be physically derived. Finally, the environmental impact is regarded as the third goal, embedded in the original time-cost optimization model,

using the atmospheric carrying capacity as a hard constraint, correcting the upper limit of rocket launch frequency, and screening the three-dimensional Pareto optimal solution.

1.4 Our Work

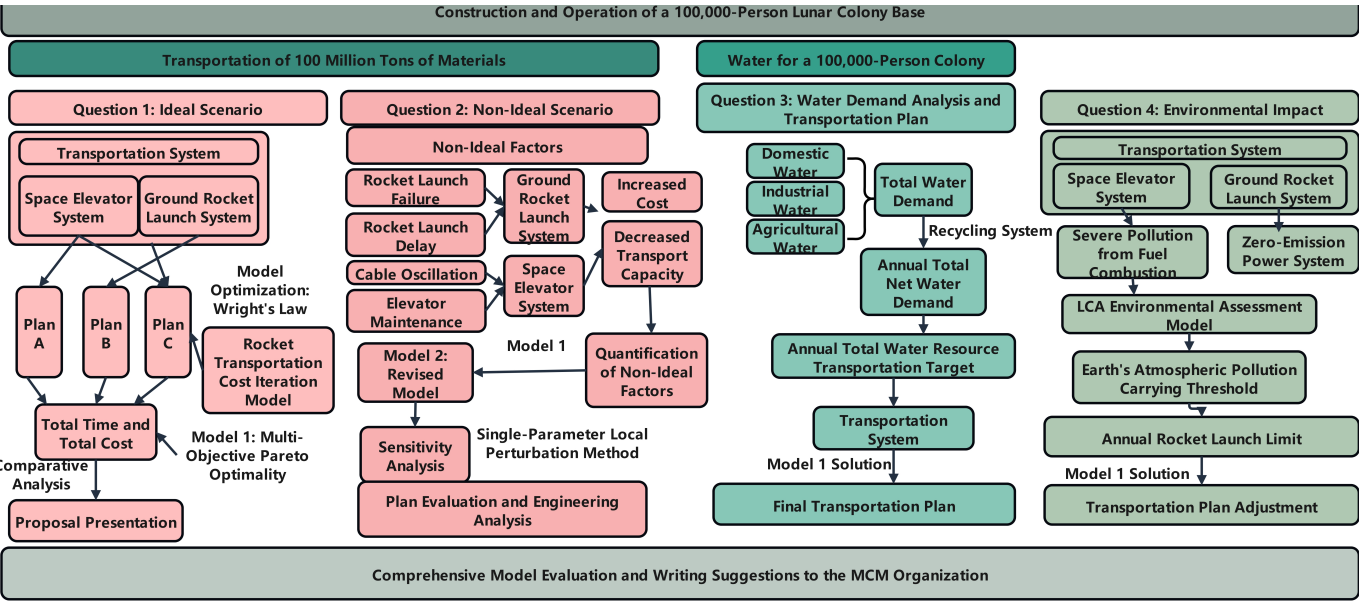


Figure 1: Our Work

2 Assumptions

1. **Assumption:** The lunar base construction and material transportation process are planned with "year" as the basic time unit. Within a single year, transportation capacity, transportation costs and resource allocation strategies remain unchanged, and transportation decisions between different years are independent of each other.

Justification: This assumption is consistent with the industry practice of conducting overall planning on an annual cycle for large-scale aerospace projects. By reasonably discretizing the time dimension, the model can focus on the core contradiction between transportation efficiency and cost while ensuring computational feasibility. Furthermore, parameter variations between years can be characterized using a technology iteration model, thereby simplifying modeling complexity while maintaining scientific rigor and engineering feasibility.

2. **Assumption:** The total amount of materials required for lunar base construction is determined at the beginning of modeling and does not change over time throughout the construction cycle. Additional material requirements due to design adjustments or unexpected needs are not considered in the model.

Justification: This assumption is consistent with the engineering practice of completing overall demand 论证 and material planning for large-scale aerospace projects at the beginning of a mission. By fixing the total material requirements, the model can effectively avoid uncertainty interference introduced by non-transportation factors, thereby focusing on analyzing the performance differences of different transportation schemes in terms of time and cost dimensions.

3. **Assumption:** In normal operating conditions, the space elevator system has a clearly defined and stable annual maximum transportation capacity, and its annual transportation volume does not exceed the system design upper limit, without considering short-term overloading operations.

Justification: This assumption is based on the fundamental principle of "safety first" in aerospace engineering and is consistent with the engineering characteristics of large-scale precision engineering systems operating under rated conditions. The model directly adopts the technical parameters given in the problem statement as system capability constraints, thereby ensuring the reliability of model inputs and the engineering interpretability of results.

4. **Assumption:** The annual availability rate of the space elevator system and the launch success rate of rockets are independent of each other across different years, and are not directly affected by historical operating conditions.

Justification: This assumption is based on the engineering characteristics that aerospace systems usually "reset" their operating status after annual maintenance and overhaul, effectively avoiding the modeling complexity caused by time dependence. At the same time, this processing method is consistent with the parameter settings given in the question, which helps to establish a unified and fair comparison benchmark between different transportation solutions.

3 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: Notations used in this literature

Symbol	Description
x	Number of rocket launches per year
M	Total mass to be transported to space (kilogram)
T	Time to complete the mission (years)
C_{total}	Total cost of the mission (dollar)
C_{SES}	Cost per kilogram of space elevator system (dollar/kg)
C_{launch}	Cost per kilogram of rocket launch (dollar/kg)
C_{total_0}	Average total market transportation cost in the base year (current)
C_{fuel}	The theoretical fuel cost floor per 100kg of cargo
α	The annual technology iteration decay factor for manufacturing and operating costs, with a value of 0.05
P_{SES}	Annual throughput of the space elevator system (kilogram/year)
P_{launch}	Payload capacity per rocket launch (kilogram/launch)

4 Modeling and Solving of Problem 1

4.1 Calculation of each scheme

4.1.1 Scheme a: Only Space Elevator System

The space elevator system consists of three galactic ports, each of which can transport 179,000 tons of cargo per year. $P_{SES} = 3 \times 179000 = 537000$ kg/year. According to predictions from the International Space Elevator Association (ISEC), the operating cost of a space elevator can be reduced to $C_{SES} = \$100/\text{kg}$, which includes power consumption, maintenance and depreciation.), the operating cost of a space elevator can be reduced to $C_{SES} = \$100/\text{kg}$, which includes power consumption, maintenance and depreciation. Therefore, the time to use the space elevator system alone is

$$T = \frac{M}{P_{SES}}$$

Total cost of the mission 为

$$C_{total} = C_{SES} \times M$$

It is calculated that using the space elevator system alone will take about 186.22 years and the total cost will be about 10trillion.

4.1.2 Scheme b: Only Rocket Launches

Using the existing ten launch sites for high-frequency launches, each rocket launch can carry a payload of $P_{launch} = 150000\text{kg}$. launches, each rocket launch can carry a payload of $P_{launch} = 150000\text{kg}$. The cost of a single rocket is $C_{launch} = \$500/\text{kg}$. If the target construction period is set to 40 years, the number of rocket launches required each year is

$$x = \frac{M}{P_{launch} \times T}$$

Total cost of the mission is

$$C_{total} = C_{launch} \times M$$

It is calculated that if only rocket launches are used and completed in 40 years, the total cost will be about 50 trillion US dollars, but 16,667 rocket launches will be needed every year, with an average of about 46 launches per day. A single launch site needs to launch about five times a day, which far exceeds the current global launch capacity and is extremely difficult in engineering.

4.1.3 Scheme c: Combined Space Elevator and Rocket Launches

When considering the two methods of space elevator system and rocket launch, this is a multi-objective optimization problem. The goal is to minimize the mission completion time T and the total cost C_{total} . Different combinations of the two methods are achieved by adjusting the number of rocket launches x to find a balance between time and cost. To analyze the trade-off between mission completion time and total cost in a hybrid space transportation system, a Hybrid Capacity Trade-off Optimization model with explicit Pareto Mapping (HCTO-PM) is developed. The model adopts a parallel transportation system structure controlled by a single parameter. The decision variable x represents the annual number of launches of the rocket. The objective function includes the mission completion time T and the total cost C_{total} . The task completion time T is given by the following formula:

$$T = \frac{M}{P_{SES} + P_{launch} \times x}$$

The total cost C_{total} is given by the following formula:

$$C_{total}(x) = C_{SES} \cdot P_{SES} \cdot \frac{M}{P_{SES} + xP_{launch}} + C_{launch} \cdot P_{launch} \cdot x$$

The model simultaneously considers the following two conflicting optimization objectives:

$$\begin{cases} \min T(x) = \frac{M}{P_{\text{SES}} + xP_{\text{launch}}} \\ \min C(x) = C_{\text{total}}(x) \end{cases}$$

Without the assumption of explicit preference weights, this problem does not pursue a single optimal solution, but instead constructs a Pareto optimal solution set to characterize the efficiency boundary in the feasible decision space. Since the model contains only one continuous decision variable, this paper uses the parameter sweep method to perform high-resolution sampling of the feasible interval of x .

Calculate the corresponding time-cost target value and obtain the Pareto front through non-dominance relationship screening.

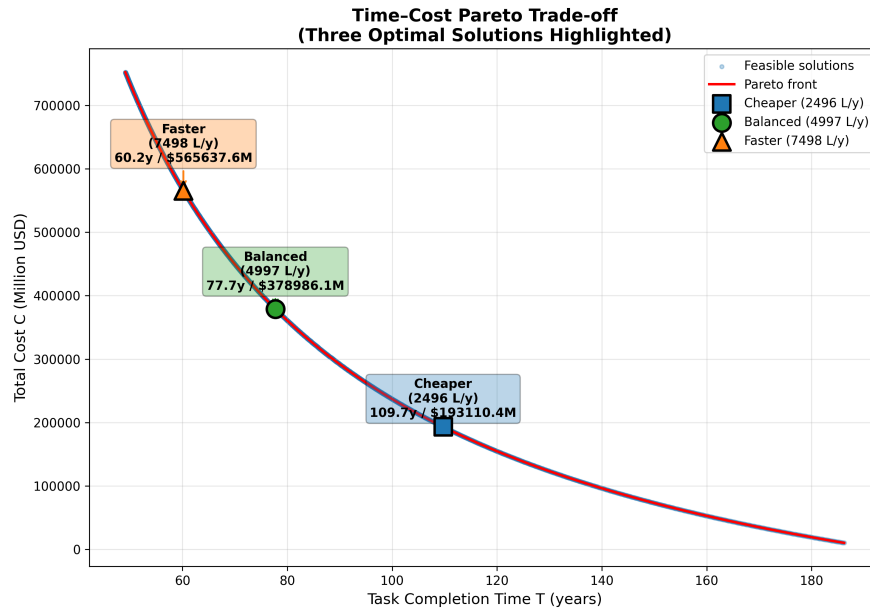


Figure 2: Pareto frontier for time-cost multi-objective optimization problems

Select some key solutions to get the following table:

Table 2: Trade-off Results under Different Rocket Launch Frequencies

Rocket Launches x (times/year)	Completion Time T (years)	Total Cost C (million USD)
2,496	109.72	193,110.40
4,997	77.72	378,986.12
7,499	49.09	752,636.23

Appropriate solutions can be selected based on different decision-making preferences.

4.2 Improvement of scheme c

4.2.1 Rocket Iteration: Transportation cost iteration

Theoretical Basis

When analyzing the evolution of transportation costs per 100kg payload, the core concept in industrial economics—the Learning Curve Model—must be introduced. That is Wright’s Law. The production cost of a unit product It does not fluctuate randomly, but shows a predictable power law downward trend as cumulative production increases. In the field of commercial aerospace, this effect is further modified into a cost iteration mechanism driven by both ”economies of scale” and ”reuse frequency”. Therefore, this study introduces the dual-factor cost structure, the unit mass cost of a single launch is deconstructed into two components with very different properties: the rocket body manufacturing and operating cost C_{body} , which is affected by the industrial learning curve, and the relatively rigid fuel (propellant) cost C_{fuel} , which is affected by chemical industry pricing. According to Wright’s Law in industrial economics, rocket body manufacturing and launch operation costs have significant ”learning by doing” characteristics, that is, as the cumulative launch volume and the number of reuses increase, this cost decreases with a power law. Therefore, the cost dynamics equation should be expressed as:

$$C_{launch} = C_{man}(n) + C_{fuel} = C_1 \cdot n^{-b} + C_{fixed}$$

Among them, n is the cumulative output, b is the learning index, and C_{fixed} represents the thermodynamic cost floor determined by chemical energy density. Model calculations show that while manufacturing costs can be rapidly amortized through a high learning rate, fuel costs constitute an insurmountable horizontal asymptote.

Parameter Settings

Based on the analysis of existing technical paths, this model sets the following core parameters:

- Manufacturing cost decay rate: Taking into account the scale effect and the maturity of reuse technology, the average annual decay coefficient α of the rocket body manufacturing and operation part is set to 5%.
- Fuel cost rigidity: The fuel cost shared per 100kg payload is set to a constant C_{fuel} as the convergence limit of cost evolution. load is set to a constant C_{fuel} as the convergence limit of cost evolution. load is set to a constant C_{fuel} as the convergence limit of cost evolution.

Taken together, future unit transportation costs will no longer approach zero infinitely, but will approach the bottom line of fuel costs along an exponential curve. This processing method suppresses the distortion risk of pure mathematical models in long-term predictions and is consistent with the physical reality of large-scale logistics networks.

Model Construction

Based on the above two-factor analysis, this study establishes the final iterative formula for the evolution of cargo transportation cost $C_{launch}(t)$ per 100kg over time t : Let t be the year (taking the model starting year as $t = 0$), $C_{launch}(t)$ consists of the variable manufacturing part $C_{man}(t)$ and the fixed fuel part C_{fuel} , then there is:

$$C_{launch}(t) = C_{man_0} \times (1 - \alpha)^t + C_{fuel}$$

Bringing in the parameters:ng in the parameters:ng in the parameters:

$$C_{launch}(t) = (C_{total_0} - C_{fuel}) \times (0.95)^t + C_{fuel}$$

This formula clearly describes the two stages of cost decline: in the near term (when t is small), it is dominated by $(1 - \alpha)^t$, which is reflected in the rapid decline of costs, consistent with the exponential characteristics of Moore’s Law; in the long term (when t tends to infinity), $C_{launch}(t)$ asymptotically converges to C_{fuel} , reflecting the ultimate constraint of energy prices on the economics of space logistics.

4.2.2 A new strategy based on the iterative models

Based on the above iterative model, we propose a new hybrid transportation strategy to further optimize the trade-off between transportation costs and time. Building upon the HCTO-PM model, we introduce a dynamic cost adjustment mechanism by incorporating the time variable t to reflect the evolution of rocket launch costs over time, resulting in Multi-Year Hybrid Capacity Trade-off Optimization with Pareto Mapping (MY-HCTO-PM) model.

Calculation formulas as follow:

$$P_{SES}T + P_{launch} \cdot \sum_{t=0}^T x(t) = M$$

$$C_{total} = C_{SES} \cdot P_{SES} \cdot T + \sum_{t=0}^T C_{launch}(t) \cdot x(t) \cdot P_{launch}$$

For each feasible T , we first calculate the total mass transported by rockets and then allocate it in reverse chronological order (starting from the last year) to take advantage of manufacturing cost decay benefits. This generates a set of (T, C) data points, i.e., the Pareto frontier. To facilitate decision-making, we define three representative solutions:

- **Balanced solution:** Use the normalized time and cost to calculate the Euclidean distance from each feasible point to the ideal point (T_{min}, C_{min}) . The one with the smallest distance is the balanced solution.
- **Radical solution (Faster):** The equilibrium solution shifts to the left by a certain proportion (reduces the total construction period and allows the cost to increase)
- **Economic solution (Cheaper):** The equilibrium solution shifts to the right by a certain proportion (extends the total construction period to reduce costs)

This method ensures that all three types of solutions fall within the optimal compromise range, and can intuitively demonstrate the trade-off relationship between construction period and cost.

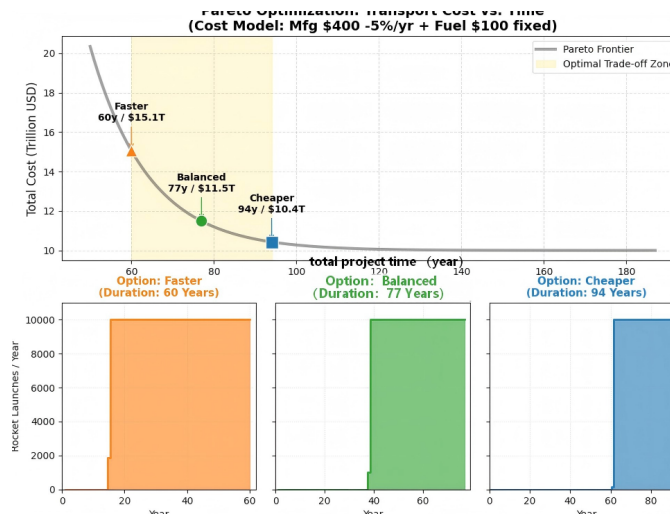


Figure 3: Multi-Year Pareto Optimization Results with Rocket Cost Iteration

4.3 Conclusions of Problem 1

The combination of space elevators and traditional rockets is more conducive to efficient and economical space transportation. By adjusting the frequency of rocket launches, the best balance between mission completion time and total cost can be found. Based on the MY-HCTO-PM model, decision makers can choose balanced, radical or economical transportation solutions based on different preferences to better meet actual needs.

5 Modeling and Solving of Problem 2

In the previous question, this article only discussed the transportation solution obtained by the elevator transportation system and the traditional rocket transportation system under ideal circumstances, individually and jointly undertaking transportation tasks. Considering the more realistic transportation conditions in reality, there are non-ideal conditions such as cable shaking, elevator damage, rocket failure and launch failure, which lead to a decrease in transportation system capacity and an increase in costs. In order to make the model more realistic, this section will discuss the transportation plan and reliability of the 100 million tons of materials required to build a 100,000-person lunar colony base under non-ideal conditions.

5.1 Introduction of Non-ideal Factors

The core of non-ideal factors is "system failure leading to capacity loss, time extension, and cost increase." It is necessary to combine aerospace engineering practice and equipment reliability rules to transform abstract faults into quantifiable parameters to ensure the operability and scientificity of the model.

5.1.1 Non-ideal Factors in Space Elevator System

The non-ideal factors of the space elevator system mainly stem from the dynamic characteristics and mechanical system losses of the ultra-long cable (100,000 kilometers of graphene cable). The specific quantification is as follows: only stem from the dynamic characteristics and mechanical system losses of the ultra-long cable (100,000 kilometers of graphene cable). The specific quantification is as follows:

Cable shaking: Affected by the disturbance of the earth's orbit and the resistance of the upper atmosphere, the cable will produce low-frequency shaking (amplitude is about 5-10 kilometers). In order to avoid the cable breakage, the transportation load needs to be reduced to 15%, thus defining the transportation capacity loss coefficient as $k_{SES} = 0.85$;

Elevator failure: Mechanical transmission system wear and tear, power supply interruption and other faults lead to system shutdown and maintenance. Referring to the reliability data of large engineering equipment, the annual availability rate is defined as $r_{SES} = 0.9$;

Repair cost: Fault repair involves cable inspection, mechanical component replacement, etc. The annual maintenance cost accounts for 5% of the annual transportation cost under perfect working conditions.

5.1.2 Non-ideal Factors in Rocket Transportation Systems

Traditional rocket transportation non-ideal factors mainly stem from the complexity of the launch process and environmental dependencies, with specific quantification as follows:

Launch failure: The structural complexity of heavy rockets (such as advanced Falcon Heavy) is high. Referring to historical data on space launches, the launch success rate is defined as $p_{launch} = 0.95$;

Launch site efficiency decline: Launch sites are affected by equipment maintenance and weather conditions (such as typhoons, thunderstorms). The launch site availability rate is defined as $r_{launch} = 0.85$, which leads to a reduction in the annual maximum number of launches at the launch site [1].

Additional costs: After a rocket launch fails, the lost material requires the rocket to be relaunched to meet the load capacity requirement. The increase in total rocket carrying capacity and rocket replenishment bring new costs, ignoring the rocket construction cost caused by rocket damage.

5.2 Transportation Plan Adjustment

5.2.1 Scheme a

- Annual transportation volume: $P_{SES,imperfect} = P_{SES} \times k_{SES} \times r_{SES}$
- Annual transportation costs: transportation costs: $C_{SES,imperfect} = C_{SES} \times P_{SES,imperfect} \times (1 + 0.05)$

Therefore, the system time using only the space elevator is $T = \frac{M}{P_{SES,imperfect}}$

Total task cost: $C_{total} = C_{SES,imperfect} \times T$

Calculated: It would take about 243.4 years to use the space elevator system alone, with a total cost of \$10.5 trillion. Compared with the ideal situation, the time increases by 30.7% and the cost increases by 5%.

5.2.2 Scheme b

- Annual launch limit: $N_{imperfect} = N \times r_{launch}$
- Total annual transportation volume: $P_{launch,total} = P_{launch} \times N_{imperfect}$
- Annual effective transportation volume: $P_{launch,imperfect} = P_{launch,total} \times p_{launch}$
- Annual transportation cost: $C_{launch,imperfect} = C_{launch} \times P_{launch,total}$

Therefore, the system time using only the rocket transportation system is $T = \frac{M}{P_{launch,imperfect}}$, and the total task cost is $C_{total} = C_{launch,imperfect} \times T$.

The calculated results show that using only the rocket transportation system requires about 82.6 years to complete the task under reasonable physical limits of launch sites, with a total cost of 52.63 trillion USD. Compared with the ideal situation, the time increase is mainly due to reduced transport capacity caused by launch site availability decline, while cost increase is mainly due to material replenishment costs caused by launch failures.

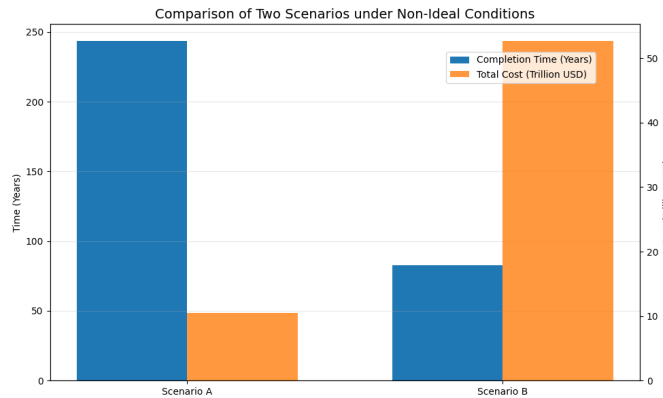


Figure 4: Comparison of Two Scenarios under Non-ideal Conditions

5.2.3 Scheme c

Comprehensively consider the two methods of space elevator system and rocket launch, continue to comprehensively consider the two methods of space elevator system and rocket launch, continue the multi-objective optimization idea in question 1, introduce non-ideal factors to correct the capacity and cost of the two systems, build an improved model, and achieve the optimal trade-off between time and cost.

Build the following model:

Decision variables:ion variables:

Number of rocket launches per year $x(t)$ (Dynamically adjusted over time, the maximum annual number of launches under non-ideal conditions is 8500, derived from 10 launch sites $\times$ 850 times/year/launch site).

Constraints:

Total capacity constraints: $P_{\text{SES,imperfect}} \times T + P_{\text{launch}} \times p_{\text{launch}} \times \sum_{t=0}^T x(t) = M$ in, $P_{\text{SES,imperfect}} = 410805$ tons/year (annual capacity of space elevator under non-ideal conditions), $p_{\text{launch}} = 0.95$ (rocket launch success rate) to ensure that the total transportation volume meets the demand of 100 million tons.

Objective function:

$$\begin{cases} T(x) = \frac{M - P_{\text{launch}} \times p_{\text{launch}} \times \sum_{t=0}^T x(t)}{P_{\text{SES,imperfect}}} \\ C(x) = C_{\text{SES,imperfect}} \times P_{\text{SES,imperfect}} \times T + \sum_{t=0}^T P_{\text{launch}} \times x(t) \times C_{\text{launch}}(t) \end{cases}$$

Among them, $C_{\text{launch}}(t)$ is the unit transportation cost of the rocket considering technology iteration (decays over time, refer to Wright's law to set the annual attenuation coefficient $\alpha = 5\%$), $C_{\text{SES,imperfect}} = 105$ USD/ton (the unit cost of the space elevator including maintenance cost bonus).

Optimize results:

High-resolution sampling of the feasible region is carried out through the parameter scanning method, and the Pareto front is obtained by filtering the non-dominated relationship, from which three types of representative solutions (radical solutions, equilibrium solutions, and economic solutions) are selected. The specific results are shown in the following table:

Table 3: Trade-off Results of Different Transportation Strategies

Option	Completion Time T (years)	Total Cost C (trillion USD)	Maximum Rocket Launches (per year)
Faster (Aggressive Strategy)	71	15.62	8500
Balanced (Compromise Strategy)	93	11.61	8500
Cheaper (Cost-Oriented Strategy)	115	10.75	8500

Based on the rocket transportation cost reduction model, launching multiple rockets in the later period has obvious economic advantages. Therefore, all plans are still to launch rockets with full load and high frequency in the later period, and launch less or no rockets in the early and mid-term. Compared with the ideal situation, the specific arrangements of plan c are still: **Galaxy Port has maintained full load and transportation for a long time, and the rocket transportation system has few or no launches in the early stage; when the cost of rockets drops, it will begin to maintain full load and high-frequency launches.**

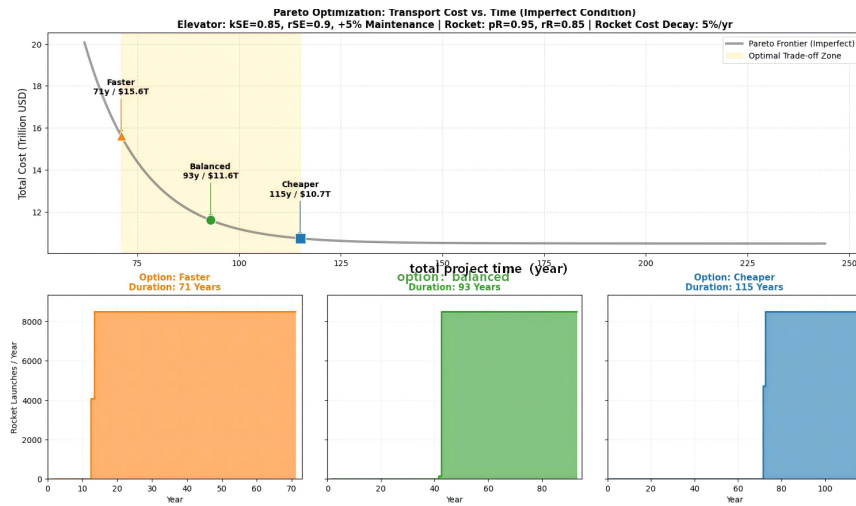


Figure 5: Pareto Optimization Results under Non-ideal Conditions

5.3 Scheme Comparison and Sensitivity Analysis

5.3.1 Comparison

In order to study the interference resistance and feasibility of the three schemes under non-ideal conditions, this article conducted a sensitivity analysis on the three scheme models and obtained the comprehensive stability score of the model. Using **single parameter local perturbation method**, for the three options a (pure space elevator), b (pure rocket), and c (mixed equilibrium solution), 6 non-ideal parameters were selected to fluctuate by $\pm 20\%$. After independent perturbation of each parameter, the change rate of time and cost was calculated. The average maximum change rate was converted into a 100-point stability score to quantify the anti-interference ability of each scheme against parameter fluctuations, providing an intuitive reference for program risk decisions.

Based on the core indicators of the three types of transportation solutions, a comparative analysis table is constructed as follows:

Table 4: Performance and Sensitivity Comparison of Transportation Schemes

Scheme	Completion Time T (years)	Total Cost C (trillion USD)	Time Sensitivity (%)	Cost Sensitivity (%)	Overall Stability Score
Pure Space Elevator	243.4	10.50	7.7	0.2	96.1
Pure Rocket	82.6	52.63	6.8	5.4	93.9
Hybrid Transportation (Scheme C, Balanced)	93	11.61	7.3	3.2	94.8

Non-ideal factors have a more significant negative impact on a single plan: the pure space elevator construction period is extended by 30.7%, the pure rocket plan has a significant increase in execution time due to launch failure and reduced launch site efficiency, and at the same time, the cost further increases. Excessively high transportation costs and long planning cycles make this plan difficult to implement.

Among the hybrid solutions, the balanced deployment of the hybrid solution is the option with the most practical application value. Its construction period is only 38.2% of that of a pure space elevator, and its cost is only 22.1% of that of a pure rocket. It combines the advantages of both solutions.

5.3.2 Engineering Measures Based on Model Results

Combining the effects of non-ideal factors and optimization results, the following engineering practice suggestions are put forward:

Space elevator system optimization:

1. In response to the problem of cable swaying, an active stability control system (such as a small thruster and a tension adjustment device) is developed to reduce the capacity loss and bring the coefficient k_{SES} to more than 0.95, which can shorten the construction period of a pure space elevator;
2. Establish a preventive maintenance system, predict failures based on equipment operation data, and increase the annual availability rate r_{SES} to 0.95, further reducing the construction period and maintenance costs.

Rocket transportation system optimization:

1. Improve rocket recovery and reuse technology to increase launch success rate p_{launch} , thereby reducing reshipment costs; at the same time, optimize the launch site scheduling system to increase launch site availability r_{launch} and reduce rocket launch costs;
2. Control the annual launch frequency of rockets within 8,500 times to avoid overdrafting the life of the launch site and balance short-term capacity and long-term sustainability.

6 Modeling and Solving of Problem 3

6.1 Preparation of models

This chapter aims to construct a multi-dimensional dynamic mass balance model (DMS model) to systematically evaluate the total water demand of a lunar colony with a population of 100,000 in full operation for one year. Due to the extreme vacuum, microgravity (1/6g) and huge temperature difference between day and night in the lunar environment, water resource management will present system dynamics that are completely different from those on the Earth [2].

The model will take into account human metabolism, intensive controlled ecological agriculture, industrial utilities and inevitable technological leaks. By extrapolating the existing Environmental Control and Life Support System (ECLSS) specifications of the International Space Station (ISS) and combining them with data from closed ecosystem experiments such as Biosphere 2 and Lunar Palace 1, the report will quantify the annual water deficit produced by a lunar colony within the constraints of closed-loop recycling efficiency, thereby determining the minimum mass of supplies that must be transported from Earth to the Moon each year. According to the previous simulation calculation based on Pareto optimal equilibrium solution, The infrastructure task is expected to reach the predetermined target node in the 77th year. Starting from the 78th year, the system will enter an operation stage with water replenishment as its core task, and a certain amount of rigid replenishment needs will be maintained every year. Based on the constructed comprehensive transportation cost model, a quantitative comparative study was conducted on three transportation efficiency paths in the replenishment stage.

6.2 Water Demand Model

6.2.1 Physiological and Domestic Water Demand

Water Flux Analysis for Maintaining Survival and Quality of Life

Based on the experience of the International Space Station, the daily basic physiological water consumption of each astronaut (including drinking, food rehydration and basic hygiene) is about 3.8 liters (1 gallon) [4]. However, for a long-term operating colony, in order to maintain the mental health and social

functions of residents, domestic water standards must be closer to the normal state of the earth. The average daily indoor water consumption per person in the average American household is about 227 to 375 liters, covering showering, laundry and toilet flushing. In the lunar environment, the circulating flux of domestic water W_{dom} can be expressed by the following formula:

$$W_{dom} = P \times (q_{drink} + q_{hygiene} + q_{laundry} + q_{food})$$

Among them, $P = 100000$, q_i represents the per capita daily demand coefficient of each sub-item. This study sets the "lunar colonization standard" per capita daily demand at 35 liters, which is much higher than the maintenance level of the ISS, but only one-tenth of the wasteful consumption on earth.

Efficiency Limitations of Water Recycling Systems (WRS)

Although the circulation flux is huge, the actual recharge requirements depend on the net efficiency η_{WRS} of the water recovery system (WRS). Currently, ISS has achieved the milestone of 98% water recovery rate through urine processor (UPA) and brine processor (BPA) [3]. Therefore, based on the current state-of-the-art recovery rate, we can estimate that by 2050, the recovery rate of the water recovery system usable in lunar colonies will reach 99%. The daily deficit of domestic water L_{dom} satisfies:

$$L_{dom} = W_{dom} \times (1 - \eta_{WRS})$$

Based on the recovery rate of 99% expected to be achieved after the lunar base is fully completed, a population of 100,000 will lose approximately 35,000 liters of water per day in the life cycle. These losses are mainly due to the saturated replacement of the filter bed, irreversible binding during the catalytic oxidation process, and unextractable bound water in the solid residue of excreta [5].

Table 5: Comparison of Domestic Water Usage and Recycling Efficiency under Different Environments

Parameter	ISS	Lunar Colony	Earth Baseline
Drinking water per capita (L/day)	2.0	3.5	3.0
Personal hygiene (L/day)	0.5	15.0	95.0
Laundry and cleaning (L/day)	0.0	10.0	70.0
Food preparation (L/day)	0.8	1.5	10.0
Toilet flushing (L/day)	0.5	5.0	100.0
Total circulation demand (L/day)	3.8	35.0	278.0
Recycling efficiency (η)	98%	99%	< 5%

6.2.2 Agricultural Water Demand

Water Footprint of Controlled Environmental Agriculture (CEA)

A colony with a population of 100,000 cannot rely on dehydrated food transported from the earth for a long time, and a large-scale lunar greenhouse must be established. The Bioregenerative Life Support System (BLSS) provides food and oxygen through photosynthesis of plants and participates in the water cycle as a natural condensed water collector. According to Biosphere 2 and NASA research, the agricultural area required to provide 100% food self-sufficiency is approximately 370 square meters per person [6]. For a population of 100,000, the total area amounts to 37 million square meters. In a hydroponic or aeroponic system, although it saves 90% of water than soil agriculture, due to the high transpiration rate of the plants,

the amount of water circulated within the system is staggering. The agricultural irrigation intensity q_{irr} is usually about 3.8 liters per square meter per day. Then the total agricultural cycle flux W_{agri_flux} is:

$$W_{agri_flux} = A_{total} \times q_{irr} \approx 1.4 \times 10^8 \text{L/day}$$

Balance Between Transpiration and Humidity Recovery

In controlled environment agriculture, most of the water absorbed by plants is released into the air through leaf transpiration. In order to close the water cycle, this humidity condensate must be recovered through a condensation heat exchanger. The net loss rate $(1 - \eta_{agri})$ of the agricultural cycle is usually much lower than that of the domestic water cycle because most of the water flows in the pure physical transpiration state. However, losses arise from three main areas:

1. Biomass fixed water, that is, the water contained in the plant tissue itself, is transferred to human metabolic circuits during harvest and consumption;
2. Regular emptying and cleaning of root solutions (due to mineral accumulation);
3. Leaks during greenhouse maintenance.

The recovery rate of the high-efficiency agricultural system is set to 99.7%, and the average daily agricultural loss is:

$$L_{agri} = W_{agri_flux} \times 0.003 \approx 420,000 \text{L/day}$$

6.2.3 Industrial Water Demand

The detailed information on industrial water use obtained through data research is shown in the table below.

Table 6: Industrial and Public Utility Water Consumption Indicators

Water Use Category	Unit Water Consumption	Estimated Daily Demand (L/day)	Key Technical Drivers
Oxygen production	1.125 kg H ₂ O per kg O ₂	94,500	Sabatier reactor water recovery efficiency
Industrial thermal control	20 L per kg H ₂	250,000	Evaporative cooling loss coefficient
Power system cooling	Fixed circulation loss ratio	100,000	Heat exchanger sealing performance
Propellant synthesis	9 L per kg H ₂ (chemical processing requirement)	150,000	Mission frequency dependency

6.2.4 Technical Leaks and Environmental Damage

Airlock Recirculation and Extravehicular Activity (EVA)

In a city of 100,000 inhabitants, frequent opening and closing of airlocks is the norm. About 4% of the air remains in each cycle and escapes to vacuum. Since the air is maintained at a relative humidity of about 50%, these escaping gases carry a non-negligible amount of water vapor. The airlock daily deficit L_{leak_air} can be modeled as:

$$L_{leak_air} = N_{EVA} \times V_{lock} \times \rho_{air} \times \omega \times (1 - \eta_{reclaim})$$

Where N_{EVA} is the number of daily cycles, V_{lock} is the airlock volume, and ω is the absolute humidity.

Structural Background Leakage

The vast surface area of lunar habitats results in tiny structural penetrations. Even if it is built to the highest aerospace standards, the annual gas loss rate per cubic meter of volume may still reach 0.01% [8]. For a colony with a volume of 3 million cubic meters, the annual water content loss due to this background leakage is constant.

6.2.5 Comprehensive Calculation

Based on the above sub-analysis, the calculation formula of the total annual supply demand M_{supply} of the lunar colony is as follows:

$$M_{supply} = 365 \times [L_{dom} + L_{agri} + L_{ind} + L_{leak_{air}} + L_{leak_{struc}}]$$

Substitute the parameter values determined in this study:

1. Loss of Life (L_{dom}): $100000 \times 35 \times 0.01 = 35000$ l/day.
2. Agricultural Losses (L_{agri}): $140000000 \times 0.03 = 420,000$ l/day.
3. Industrial Losses (L_{ind}): Estimated to be 500,000 l/day.
4. Technical Leak ($L_{leak_{air}} + L_{leak_{struc}}$): Including EVA and structural leakage, estimated at 15,000 l/day.

Table 7: Annual Water Mass Balance Summary (100,000-Person Scale)

Loss Category	Daily Loss (L/day)	Annual Loss (Metric Tons)	Share of Total (%)
Domestic water circulation	35,000	12,775	3.6
Agricultural BLSS system	420,000	153,300	43.3
Industrial use, oxygen production and cooling	500,000	182,500	51.5
Airlock operations, leakage and EVA	15,000	5,475	1.5
Total	987,500	354,050	100.0

According to model calculations, within one year of full operation of a lunar colony with a population of 100,000, the total water consumption will be approximately 354 million liters. Under ideal recovery rates in all parts, a net water resource deficit of approximately 354,000 metric tons per year would have to be faced. Without the intervention of in-situ resource utilization (ISRU), nearly 1,000 tons of water would need to be moved from the earth to the lunar surface every day on average.

6.3 Water Resource Replenishment Model

According to previous simulation calculations based on Pareto optimal equilibrium solutions, the infrastructure task is expected to reach the predetermined target node in the 77th year. Starting from the 78th year, the system will enter an operation phase with water replenishment as its core mission, and will need to maintain a rigid replenishment demand of 354,200 metric tons per year.

Pure space elevator system transportation solution shows extremely high economic stability. The model shows that the annual design capacity limit of the space elevator is 5.37×10^5 tons, and the capacity

utilization rate remains stable at 65.96%, which can fully cover people's livelihood supply needs. Its unit transportation cost is anchored at 100 USD/kg, and the total annual cost is approximately 35.42 billion USD. With its nonlinear cost advantage and electric energy drive characteristics, this system is the most sustainable economic form to maintain the operation of the "artery" of earth-lunar logistics.

Pure Rocket Cluster System Transportation Solution Limited by chemical energy and physical properties, the total unit transportation cost of the rocket cluster is 116.6 USD/kg (including a fixed fuel cost bottom line of 100 USD/kg), and the total annual cost is approximately US\$41.30 billion, which is US\$5.88 billion higher than the elevator solution. Although it has the flexibility of discrete deployment, its economic efficiency, environmental load and launch site turnover pressure are significantly inferior to orbital transportation in ultra-large-scale, long-term standardized material supply.

Dual system comprehensive scheme Based on the MY-HCTO-PM model logic, this scheme adopts the principle of "marginal cost minimization". During regular replenishment periods, the system automatically allocates 100% load to the elevator system to achieve minimum expenditure (\$35.42 billion). Its core strategic value lies in the construction of a robust redundancy mechanism of "fixed orbit main transportation + discrete mobile backup": retaining the potential of 10,000 rocket launches per year, ensuring that when the elevator system is out of service due to force majeure, the logistics chain has the supply resilience to immediately intervene.

6.4 Sensitivity Analysis and Resource Utilization

Model results show that agricultural and industrial losses account for more than 94% of total recharge needs. This reveals a key insight: increasing the recycling rate of domestic water (for example from 98% to 99%) will have a minimal marginal contribution to reducing logistics pressure. Conversely, a 0.1% improvement in agricultural condensation efficiency could save tens of thousands of tons of supply mass per year.

6.5 Conclusions of Problem 3

This report demonstrates the huge logistical challenges in the early stages of operating a lunar colony with a population of 100,000 by building a cross-domain mathematical model. Even with state-of-the-art closed-loop recycling technology, the annual water replenishment demand of 354,000 metric tons still exceeds current and near-future (such as Starship cluster) transportation capabilities. Therefore, full operation of the lunar colony must be synchronized with in-situ water resource extraction on the lunar surface.

To sum up, the best strategy for water replenishment of the lunar base should be a comprehensive transportation paradigm of "elevators as the main body and rockets as elastic backup". In the long-term operation launched in the 78th year, this model not only achieved annual cost savings of approximately 14.2% by maximizing asset utilization, but also utilized the elevator's annual capacity surplus of approximately 182,800 tons to reserve logistics redundancy for the base's future scale expansion.

7 Modeling and Solving of Problem 4

7.1 Preparation of models

Traditional cost-time optimization models often ignore the cumulative effect of high-frequency heavy rocket launches on the stratosphere and global climate. In order to evaluate the impact of the monthly material transportation demand of up to 10^8 tons anywhere on the earth's atmospheric environment, this chapter introduces environmental impact factors based on life cycle assessment (LCA) on the basis of the original logistics optimization model, and establishes a threshold constraint mechanism based on the atmospheric carrying capacity, aiming to quantitatively analyze the ecological costs of different transportation strategies.

7.2 Construction of LCA Model

In traditional surface transportation LCA analysis, environmental impact is usually mainly characterized by carbon dioxide CO_2 emissions. However, space transportation has its own uniqueness: rocket

engines directly inject combustion products into the stratosphere and mesosphere. Due to the lack of convective weather phenomena (such as rainfall) at these altitudes, the residence time of particulate matter can last for years. Therefore, the single launch environmental impact equivalent (μ) defined in this model is a composite indicator, and its calculation formula is as follows:

$$\mu = \alpha \cdot M_{CO_2} + \beta \cdot M_{BC} + \gamma \cdot M_{Al_2O_3} + \delta \cdot M_{H_2O}$$

Where:

- M_{CO_2} is the greenhouse gas emission mass.
- M_{BC} is the mass of black carbon, soot particles.
- $M_{Al_2O_3}$ is the mass of aluminum oxide particles (mainly from solid boosters).
- M_{H_2O} is the water vapor in the upper atmosphere.
- $\alpha, \beta, \gamma, \delta$ are the environmental destruction weight coefficients for each pollutant.

In the numerical simulation, we simplified the above complex physical and chemical processes into a comprehensive scalar $ENV_{IMPACT_PER_LAUNCH}$ (set to 2500 EIU), which represents the comprehensive marginal damage to the global environment caused by the launch of a heavy-lift launch vehicle (such as Starship class).

7.3 Calculation of Total Environmental Impact

7.3.1 Physical Derivation of Atmospheric Carrying Capacity Threshold

The total environmental impact E_{total} of the system is a function of time T . Since the space elevator relies mainly on electricity during its operation phase (assumed to be powered by clean energy), its direct environmental emissions are considered zero (or negligible relative to rockets). Therefore, the total environmental impact is dominated by rocket launch activities. Assuming L_t is the number of rocket launches in year t , the total environmental load of the system is calculated as follows:

$$E_{total}(T) = \sum_{t=1}^T (L_t \times \mu)$$

Where L_t is endogenously derived from the economic dispatch strategy determined by the current year's remaining transportation demand $M_{remain}(t)$ and rocket manufacturing/fuel costs. This formula reveals the strong coupling relationship between environmental impact and transportation strategy: the pursuit of shortening the total construction period T will inevitably lead to a surge in L_t in the early stage, resulting in an exponential increase in E_{total} . This paper derives the maximum safe annual emissivity based on the steady-state load of stratospheric aerosols. The natural background aerosols in the Earth's stratosphere are mainly composed of sulfates, and their total mass is relatively stable.

We define the following core parameters:

- M_{bg} : Total stratospheric background aerosol mass, (6×10^8 kg).
- η : Maximum allowable perturbation ratio (Perturbation Ratio).
- τ : Average residence time of particles (Residence Time).
- m_{eff} : Effective particle injection volume in a single shot (2×10^3 kg).

7.3.2 Steady-state Equilibrium Calculation Formula

When launch activities continue for a long time, the mass M_{human} of man-made pollutants in the stratosphere will reach a dynamic balance, that is, the annual injection amount is equal to the deposition amount. The equilibrium equation is:

$$M_{human} = C_{atm} \times m_{eff} \times \tau$$

According to environmental safety constraints:

$$M_{human} \leq M_{bg} \times \eta$$

Combining the above two formulas, the calculation formula of the maximum safe annual emissivity C_{atm} can be derived:

$$C_{atm} = \frac{M_{bg} \times \eta}{m_{eff} \times \tau}$$

Substitute specific values for calculation: Calculate the maximum allowable human load increment:

$$M_{limit} = 6 \times 10^8 \text{ kg} \times 0.05 = 3 \times 10^7 \text{ kg}$$

Substitute into the equilibrium equation to solve for the number of launches:

$$C_{atm} = \frac{3 \times 10^7 \text{ kg}}{2000 \text{ kg/launch} \times 3.75 \text{ years}} = 4000 \text{ launches/year}$$

7.4 Modification of the Transportation Model

Based on this model, we concluded that 4000 times/year is the theoretical physical upper limit of the earth's atmosphere without significant climate changes. Therefore, the decision logic is revised to:

$$S_{status} = \begin{cases} \text{Sustainable (Green),} & \text{if } \max(L_t) \leq 4000, \\ \text{Environmental Hazard (Red),} & \text{if } \max(L_t) > 4000. \end{cases}$$

The model running results are shown in Figure 6:

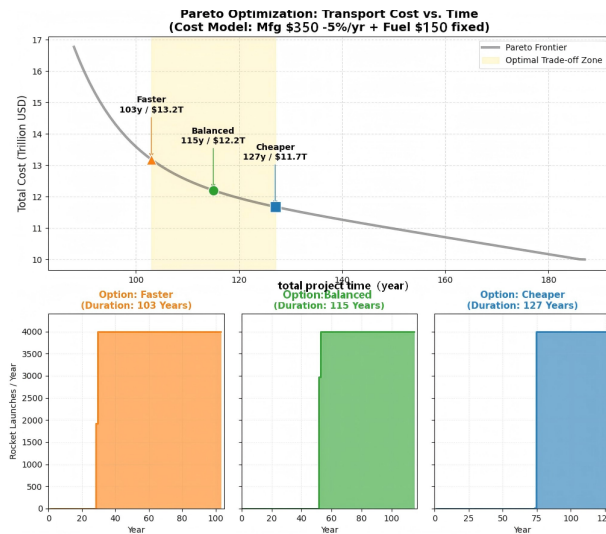


Figure 6: Modification of Pareto Optimization under Environmental Constraints

It can be seen that after correcting the model for environmental factors, the corresponding Pareto solution changes.

7.5 Conclusion

Through the construction of the above formula, this model expands the original two-dimensional "time-cost" Pareto optimization into a three-dimensional "time-cost-environment" trade-off analysis. Mathematical derivation shows that since the unit environmental cost μ of rocket transportation is constant and significantly greater than zero, while the unit environmental cost of the space elevator approaches zero, the mathematical solution to minimizing the environmental impact E_{total} must point to a solution that maximizes the utilization rate of the space elevator. This quantitatively proves that in a billion-ton Earth-moon transportation mission, extending the construction period to wait for the release of the space elevator's capacity is not only the optimal economic solution, but also the only path that the earth's ecosystem can withstand.

8 Model Evaluation

8.1 Advantages

The developed model system is systematic and forward-looking, characterized by the following three strengths:

1. **Economic and Logistics Optimization:** The MY-HCTO-PM model transcends traditional single-objective optimization by integrating a "two-factor cost structure" and Wright's Law. This framework captures the physical trajectory of declining space transportation costs, avoiding the distortions of purely mathematical forecasting. Furthermore, Pareto Mapping provides practical decision-making trade-offs between mission cycles and budgetary constraints.
2. **High-Precision Resource Management:** The water demand model for a lunar colony of 100,000 inhabitants accounts for complex fluxes—including physiological metabolism, ecological agriculture, industrial production, and technical losses such as airlock leakage. Validated against International Space Station (ISS) and closed-ecosystem experimental data, this multi-dimensional balance ensures a robust physical foundation for Earth-Moon logistics.
3. **Innovative Sustainability Framework:** By incorporating Life Cycle Assessment (LCA) and stratospheric aerosol carrying capacity, the study introduces "environmental cost" as a third dimension to the time-cost trade-off system. This elevates the framework to a sustainable development model, quantitatively demonstrating the unique ecological advantages of space elevators for ultra-large-scale transportation.

8.2 Limitations

Despite its performance, the model possesses certain limitations:

1. **Linearity in Technology Iteration:** The model relies on historical parameters (e.g., 5% annual decay), which may fail to capture the nonlinear, disruptive breakthroughs characteristic of aerospace technology. Sudden advancements in materials or propulsion ("black swan" events) are difficult to account for within the current exponential decay framework.
2. **Steady-State Assumptions:** The model simplifies the nonlinear coupling between lunar population growth and infrastructure expansion. By assuming a steady state post-construction, it may overlook the dynamic complexities of population migration and industrial ramping, potentially introducing biases during early expansion phases.

9 Letter of Advice

To: MCM Lunar Development Technical Committee

Subject: Technical Integration and Strategic Framework for the Development of a 100,000-Resident Lunar Colony

Dear Members of the Technical Committee,

In the context of the continuous advancement in aerospace engineering and deep-space exploration capabilities, the establishment of a lunar colony capable of supporting a permanent population of 100,000 has emerged as a tangible engineering challenge centered on systems engineering and the integration of critical technologies. Toward this objective, we propose the following recommendations from the perspectives of technical implementation and long-term operational sustainability.

Regarding Spatial Siting and Structural Design, the lunar south pole should be prioritized as the primary site. This region offers relatively stable illumination conditions and is rich in water ice resources within its Permanently Shadowed Regions (PSRs), which are essential for energy acquisition and life support. Residential and occupational zones should utilize modular, hermetically sealed, and semi-subterranean structures. By leveraging lunar regolith coverage for radiation shielding, thermal regulation, and micrometeoroid defense, the requirement for imported protective materials can be significantly reduced.

Regarding Life Support Systems, the colony must rely on a highly reliable closed-loop technological architecture. Water self-sufficiency can be achieved through ice mining and advanced wastewater treatment and recycling systems, aiming for a recovery rate of over 99

Energy Systems form the cornerstone of colonial operations. We recommend large-scale solar arrays as the primary energy source, supplemented by small-scale nuclear reactors for stability and emergency backup. Energy will not only sustain life support and habitats but also power industrial manufacturing, resource processing, and infrastructure expansion. This necessitates high-efficiency energy storage and intelligent distribution systems to mitigate the effects of the lunar diurnal cycle and sudden load fluctuations.

For Industrial and Infrastructure Construction, priority must be given to the development of In-Situ Resource Utilization (ISRU) technologies. Through the processing of lunar regolith, oxygen, metallic materials, and construction feedstock can be extracted for structural fabrication, equipment maintenance, and consumables production. This strategy will drastically lower long-term logistics costs and serves as the key technical pillar for large-scale, sustainable colonization.

At the Demographic and Social Operational level, population introduction should be phased, starting with scientific research, engineering, and medical personnel, followed by the gradual expansion of education, service, and administrative sectors. Comprehensive medical care, psychological support, and robust social organizational structures are as critical as technical systems for the stable survival of humans in long-term enclosed environments.

Finally, regarding Safety and Long-term Maintenance, the colony must incorporate multi-layered redundancy designs, including backup capacities for life support, energy supply, and communication systems. Systematic fault detection and emergency response mechanisms must be established to address equipment aging, environmental anomalies, and unforeseen risks.

In summary, the realization of a 100,000-person lunar colony depends on the mature integration of

multiple key technologies rather than a single breakthrough. By steadily advancing resource utilization, energy security, and life support technologies, this goal remains feasible from an engineering perspective in the long term.

Respectfully submitted,
Team 2623614

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